UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 009 · Particular integral when X is any other form · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·5	Unit 2 deck 009 ("Particular integral when X is any other form", SESSION 7), slide 5 (rendered .pptx)
T009 00:03:13	Unit 2 AV-summary transcript "Particular Integral when X is any other form" (15.2 min), segment start
T009b 00:01:10	Unit 2 AV-summary transcript "Particular integral when X is any other form_Continued" (10.2 min), segment start
CW U2 P28	Unit 2 class-work problem 28 (pointer only; solved in the class 011 VoP pack)
ESA Dec-23 3c	Past paper: UE22MA141A Dec 2023, question 3.c

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4. Past-paper questions

1 The general method: linear factors, partial fractions and $\frac{1}{D-m}X$

Definitions

"Any other form". A right-hand side X that is not one of the four standard forms: not e^{ax}, not $\sin(ax+b)$ or $\cos(ax+b)$, not a polynomial x^m, and not $e^{ax}V$. Examples: $\tan ax$, $\ln x$, e^{e^x}, $\sin(e^x)$.

S009·3 · T009 00:00:07 · T009 00:00:27 · T009b 00:00:32

Linear factors of $f(D)$. $f(D)=(D-m_1)(D-m_2)\cdots(D-m_n)$, where $m_1,\ldots,m_n$ are the roots of the auxiliary equation $f(m)=0$ (real or complex). S009·3 · T009 00:01:19

Inverse operator $\frac{1}{D-m}$. $\frac{1}{D-m}X$ is the function u with $(D-m)u = X$, i.e. $u' - mu = X$; it equals $e^{mx}\int Xe^{-mx}\,dx$ (no constant of integration: the constants are in the CF). S009·4 · T009 00:02:11 · T009b 00:03:01

$\frac{1}{D^k}$. k successive integrations with respect to x (the case $m = 0$ of $\frac{1}{D-m}$, applied k times). S009·8 · T009 00:12:09

Statements in the class material

> *"When X is any other function of x. Rules for Finding the Particular Integral. Consider the differential equation, $(D^n + a_1D^{n-1} + a_2D^{n-2} + \cdots + a_{n-1}D + a_n)y = X$. It can be written as: $f(D)y = X$ ∴ P.I. $= \frac{1}{f(D)}X$. Resolve $f(D)$ into linear factors. Let $f(D) = (D-m_1)(D-m_2)\ldots(D-m_n)$. Then, P.I. $= \frac{1}{f(D)}X = \frac{1}{(D-m_1)(D-m_2)\ldots(D-m_n)}X$."* S009·3

> *"Use partial fractions: $P.I = \left(\frac{A_1}{D-m_1} + \frac{A_2}{D-m_2} + \cdots + \frac{A_n}{D-m_n}\right)X$, $P.I = A_1\frac{1}{D-m_1}X + A_2\frac{1}{D-m_2}X + \cdots + A_n\frac{1}{D-m_n}X$, $P.I = A_1e^{m_1x}\int Xe^{-m_1x}dx + A_2e^{m_2x}\int Xe^{-m_2x}dx + \cdots + A_ne^{m_nx}\int Xe^{-m_nx}dx$."* S009·4

> *"Standard formula (lecture): "one over D minus a of $f(x)$ ... is nothing but e^{ax} integral $e^{-ax}f(x)\,dx$"."* T009 00:02:11 · T009b 00:03:01

Key points

1. First check the four standard forms (e^{ax}; $\sin/\cos(ax+b)$; x^m; $e^{ax}V$). Only if X is none of them ($\tan ax$, $\ln x$, e^{e^x}, $\sin(e^x)$) use this method. ESA ★★ T009 00:00:27 · T009 00:03:13 · T009b 00:00:32 · T009b 00:01:25
2. "There's no change in computing the complementary function": AE, roots, CF exactly as in class 002. ESA ★★ T009 00:03:33 · S009·5
3. Procedure: factorise $f(D)$ into linear factors → partial fractions $\sum\frac{A_k}{D-m_k}$ → $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$ for each term → add (Figure F1.1). ESA ★★ S009·3 · S009·4 · T009 00:01:19 · T009 00:01:52 · T009 00:02:11
4. Watch the sign: $\frac{1}{D+2}X = e^{-2x}\int Xe^{2x}dx$ ($m = -2$): the outside exponential has the AE root, the inside one its negative. ESA ★★ T009b 00:03:36 · S009·11 · S009·12
5. No constants of integration in a PI: the arbitrary constants are already in the CF. ESA ★★ (added for clarity) S009·9 · S009·11
6. With $m = 0$ ($f(D) = D^2$), $\frac{1}{D^2}$ means "integrate twice": $\frac{1}{D^2}\ln x = \frac{x^2\ln x}{2} - \frac{3x^2}{4}$. ESA ★ S009·8 · S009·9 · T009 00:12:09
7. Shortcut (not on the slides): apply the factors one after another, $\frac{1}{D-m_1}\left[\frac{1}{D-m_2}X\right]$; this avoids partial fractions and also works for repeated factors. ESA ★★ (added for clarity) S009·3 · ESA Dec-23 3c

Why?

Why does $\frac{1}{D-m}X$ equal $e^{mx}\int Xe^{-mx}dx$?
$u=\frac{1}{D-m}X$ means $u'-mu=X$, a first-order linear equation. Multiplying by the integrating factor e^{-mx} gives $(ue^{-mx})'=Xe^{-mx}$; integrating and multiplying by e^{mx} gives the formula.

ADDED FOR CLARITY S009·4

Why do the shortcut rules of Types 1–5 fail for $\tan ax$ or e^{e^x}?
Those rules work because D maps e^{ax}, $\sin ax$, x^m into the same family. The derivatives of $\tan ax$ or e^{e^x} are new functions every time, so no substitution $D\to a$ or $D^2\to -a^2$ exists; the integral formula always works.

ADDED FOR CLARITY T009 00:00:27 · T009 00:03:13

Exam traps

Exam trap. Do not write $\frac{1}{D+1}X=e^x\int Xe^{-x}dx$: for $D+1=D-(-1)$, $m=-1$, so it is $e^{-x}\int Xe^{x}dx$. ADDED FOR CLARITY T009b 00:03:36 · S009·12

Exam trap. Partial fractions with the wrong sign: $\frac{1}{(D+1)(D+2)}=\frac{1}{D+1}-\frac{1}{D+2}$ ($A=1$ at $D=-1$, $B=-1$ at $D=-2$). Slide 009·10 lists the fractions as $\frac{A}{D+2}+\frac{B}{D+1}$, so there $A=-1$, $B=1$ — the same split. S009·10 · S009·12

Exam trap. Do not forget the CF: "solve" means $y=y_c+y_p$ (slides 009·10–17 stop at the PI; the ESA question asks to solve). ADDED FOR CLARITY ESA Dec-23 3c · S009·11

2 Complex linear factors: $(D^2 + a^2)y = \tan ax$

Key points

1. $D^2 + a^2 = (D + ia)(D - ia)$, so $\frac{1}{D^2+a^2} = \frac{1}{2ia}\left[\frac{1}{D-ia} - \frac{1}{D+ia}\right]$. ESA ★★ S009·5 · S009·6 · T009 00:04:31
2. Compute $\frac{1}{D-ia}\tan ax$ once; get $\frac{1}{D+ia}\tan ax$ by changing i to $-i$ ("we simply replace i by $-i$"). ESA ★★ S009·7 · T009 00:08:58
3. The imaginary parts must cancel: the PI of a real equation is real ($e^{iax} + e^{-iax} = 2\cos ax$ at the end). ESA ★★ S009·7 · T009 00:10:30 · T009 00:10:48
4. Result to remember: $\frac{1}{D^2+a^2}\tan ax = -\frac{1}{a^2}\cos ax \log(\sec ax + \tan ax)$. ESA ★★ S009·7 · T009 00:11:09
5. $\tan ax$, $\sec ax$, $\csc ax$ right-hand sides can equally be solved by variation of parameters (class 011); the CW sheet asks them that way (P23, P26, P28). Read the question: "by the method of variation of parameters" forces that method. ESA ★★ ADDED FOR CLARITY CW U2 P23 · CW U2 P26 · CW U2 P28

Why?

Why is the PI real although the computation is complex?
The equation has real coefficients and a real right-hand side. The two partial results are complex conjugates, so $\frac{1}{2ia}(P_1 - \bar{P}_1) = \frac{1}{a}\mathrm{Im}\, P_1$ is real.
ADDED FOR CLARITY S009·7 · T009 00:10:30

Exam traps

Exam trap. Do not integrate $\frac{\sin^2 ax}{\cos ax}$ in one go: write it as $\sec ax - \cos ax$ first, then $\int \sec ax\, dx = \frac{1}{a}\log(\sec ax + \tan ax)$. ADDED FOR CLARITY S009·6 · T009 00:06:46

Exam trap. The PI has $\frac{1}{a^2}$, not $\frac{1}{a}$: one $\frac{1}{a}$ from the integration and one from $\frac{1}{2ia}$.
ADDED FOR CLARITY S009·7

3 Right-hand sides in e^x: the substitution $t = e^x$

Definitions

Substitution $t = e^x$. $dt = e^x\,dx$, so $e^{kx}\,dx = t^{k-1}\,dt$; it turns $\int e^{kx}F(e^x)\,dx$ into $\int t^{k-1}F(t)\,dt$. S009·11 · S009·12 · S009·15 · T009b 00:03:57

Bernoulli's rule. Repeated integration by parts in one line: $\int uv\,dx = uv_1 - u'v_2 + u''v_3 - \cdots$, where $v_1, v_2, \ldots$ are successive integrals of v; used for $\int t^3 \sin t\,dt$, $\int t^4 \sin t\,dt$. T009b 00:08:37 · S009·16

Key points

1. For X built from e^x, substitute $t = e^x$, $dt = e^x dx$ in each integral: $e^{kx}dx = t^{k-1}dt$. ESA ★★ S009·11 · S009·12 · S009·15 · T009 00:14:05 · T009b 00:03:57
2. Split the exponential to make $e^x dx$ visible: $e^{2x} = e^x \cdot e^x$, $e^{4x} = e^{3x} \cdot e^x$ ("let us split e^{2x} to be e^x times e^x"). ESA ★★ T009 00:14:22 · T009b 00:04:57 · T009b 00:08:00
3. ESA Dec-23 3c ($y'' + 3y' + 2y = e^{e^x}$, 7 marks) is the slide 009·10 problem: PI $= e^{-2x}e^{e^x}$; write the full y. ESA ★★ ESA Dec-23 3c · S009·10 · S009·11
4. Always simplify after substituting back: in $\frac{1}{(D+1)(D+2)}\sin(e^x)$ the terms $-e^{-x}\cos(e^x)$ cancel, leaving $-e^{-2x}\sin(e^x)$. ESA ★★ S009·13 · T009b 00:06:04
5. Choosing the order $\frac{1}{D+2}\left[\frac{1}{D+1}X\right]$ for e^{e^x} and $\sin(e^x)$ needs only $\int e^t dt$ and $\int \cos t\,dt$: two lines each (W3.1, W3.2). ESA ★★ ADDED FOR CLARITY S009·10 · S009·12 · ESA Dec-23 3c

Why?

Why does the order of successive factors matter for e^{e^x}?
$\frac{1}{D+1}e^{e^x} = e^{-x}e^{e^x}$ is easy, and then $\frac{1}{D+2}$ needs $\int e^{2x}e^{-x}e^{e^x}dx = \int e^x e^{e^x}dx = e^{e^x}$. The other order leaves $\int(1 - e^{-x})e^{e^x}dx$, which needs spotting a derivative.
ADDED FOR CLARITY S009·10 · ESA Dec-23 3c

Exam traps

Exam trap. Slide 009·16 (and 009·17) writes the first term of (1) as $-e^{-x}\cos(e^x)$; it is $e^{-2x} \cdot (-e^{3x}\cos(e^x)) = -e^x\cos(e^x)$. The boxed answer is right only because the correct $-e^x\cos(e^x)$ cancels with (2). ADDED FOR CLARITY S009·16 · S009·17

Exam trap. After $t = e^x$, also convert the dx: $e^{2x}e^{e^x}dx = te^t\,dt$, not $t^2e^t\,dt$ (one e^x goes into dt). ADDED FOR CLARITY S009·11 · T009 00:14:22

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). This session's method is asked exactly once: ESA Dec-23 3c, $y'' + 3y' + 2y = e^{e^x}$ (7 marks), never repeated in the later papers; it is the same equation as slide 009·10 (W3.1). Stars: partial fractions with $\frac{1}{D-m}X$ and the substitution $t = e^x$ ★★ (one ESA question); $\tan ax$ ★★ (its answer with $a = 2$ is CW P28, which the sheet asks by variation of parameters); $\frac{1}{D^2}\ln x$ and the complex-factor details ★. No CW problem is assigned to this session: the CW sheet's tan, sec, csc problems (P23, P26, P28) are in the variation-of-parameters section and are solved in the class 011 VoP pack.

3 · Right-hand sides in e^x: the substitution $t = e^x$

ESA Dec 2023 · 7 marks · UE22MA141A Q3.c · solved in W3.1

Solve $y'' + 3y' + 2y = e^{e^x}$.